

Problem Statement – LectMent: AI-Powered Smart Learning Assistant

The Challenge

With the rapid growth of online learning and digital education, students are exposed to lengthy lectures, technical presentations, and educational videos that are often difficult to understand and revise. Manual note-taking during lectures is time-consuming and frequently results in incomplete or unstructured study materials. This challenge is even greater for **non-native speakers, self-learners, and neurodivergent students**, who may struggle to keep pace with fast-paced lectures or complex subject matter.

Although general AI chatbots can answer questions, they are not specifically designed to convert educational content into structured learning resources such as study notes, revision sheets, quizzes, flashcards, and key takeaways. As a result, students spend significant time organizing lecture content instead of focusing on learning and revision.

The Need

There is a growing need for an intelligent educational assistant that can automatically analyze lecture content from multiple sources—including transcripts, YouTube lectures, and audio recordings—and transform it into concise, accurate, and personalized study materials. Such a solution should improve learning efficiency, reduce manual effort, and support students with diverse learning needs.

Proposed Solution

LectMent is an AI-powered Smart Learning Assistant built using **IBM watsonx.ai**, **IBM Cloud**, and **Agentic AI**. It accepts lecture transcripts, YouTube lecture links, and audio recordings as input, understands the educational context using Foundation Models, and automatically generates structured learning resources including **summaries, study notes, revision sheets, quizzes, flashcards, and key takeaways**. By automating lecture understanding and content organization, LectMent enables students to learn more effectively, revise faster, and improve knowledge retention.